Funduino Part 2 Project

Joe Parker, Luke Clemens and Robert Danner



PennState

Introduction

Importance of feedback design

Feedback controller design is necessary to accurately track and depict motors at desired speeds while limiting real world disturbances.

This report covers the open and closed loop behavior of a DC motor system and its implementation to Arduino.

Transfer Function

$$G(s) = \frac{K}{\tau s + 1}$$

The original transfer function in project 1 was depicted as a second order system.

Through thorough analysis and derivation, this model can be displayed as first order due to no oscillatory motion within the motor.

Open-Loop Analysis

Open Loop Transfer Function:

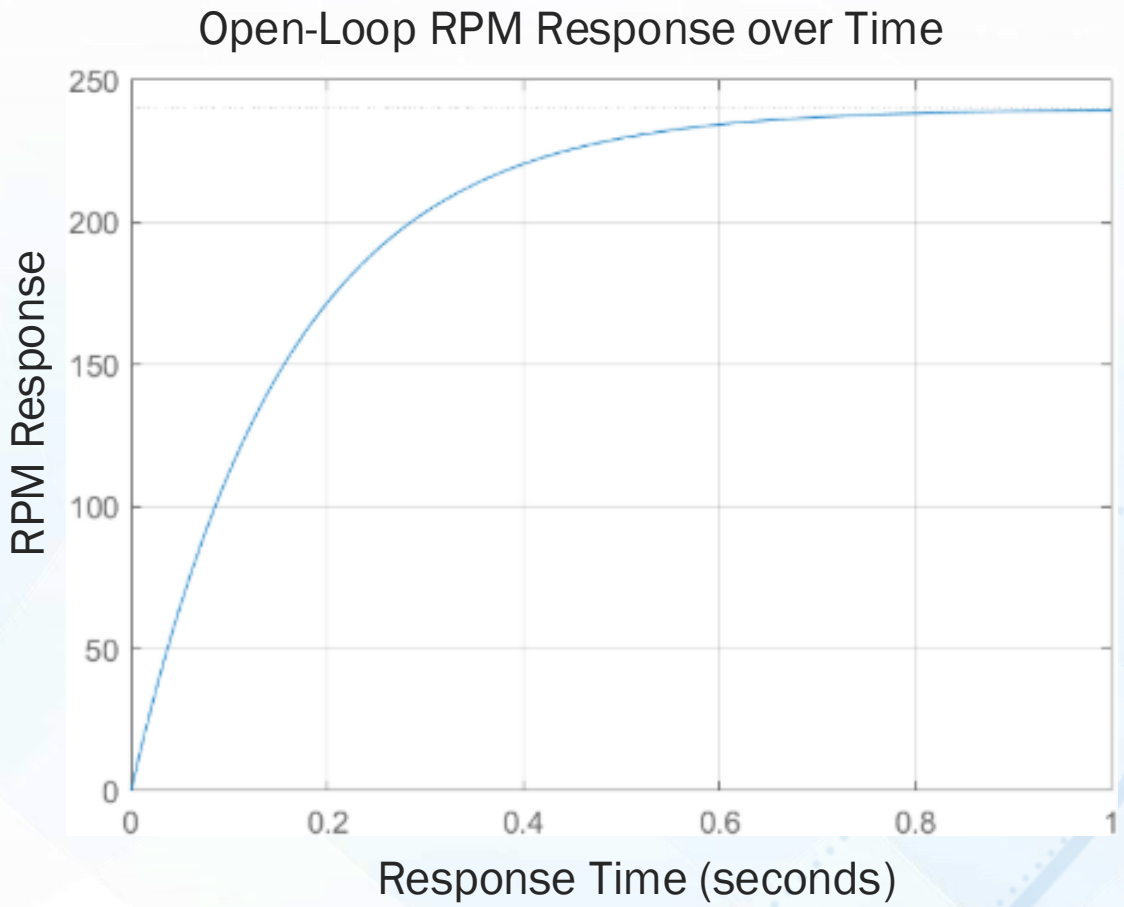
$$G(s) = \frac{31.2375}{0.159s + 1}$$

Zero: N/A

Pole: s = -6.29 rad/s

$$Y(s) = G(s)\cdot(\ 240/s\ ) = \frac{31.2375}{0.159s + 1}\cdot(\ 240/s\ )$$

Rise Time: 0.349 Seconds



Closed-Loop Analysis

Closed Loop Transfer Function with Unity Feedback:

$$T(s) = \frac{G(s)}{1 + G(s)} = \frac{31.2375}{0.159s + 32.2375}$$

Zeros: N/A
Pole: s = -202.7 rad/s

Laplace Domain Step Response:

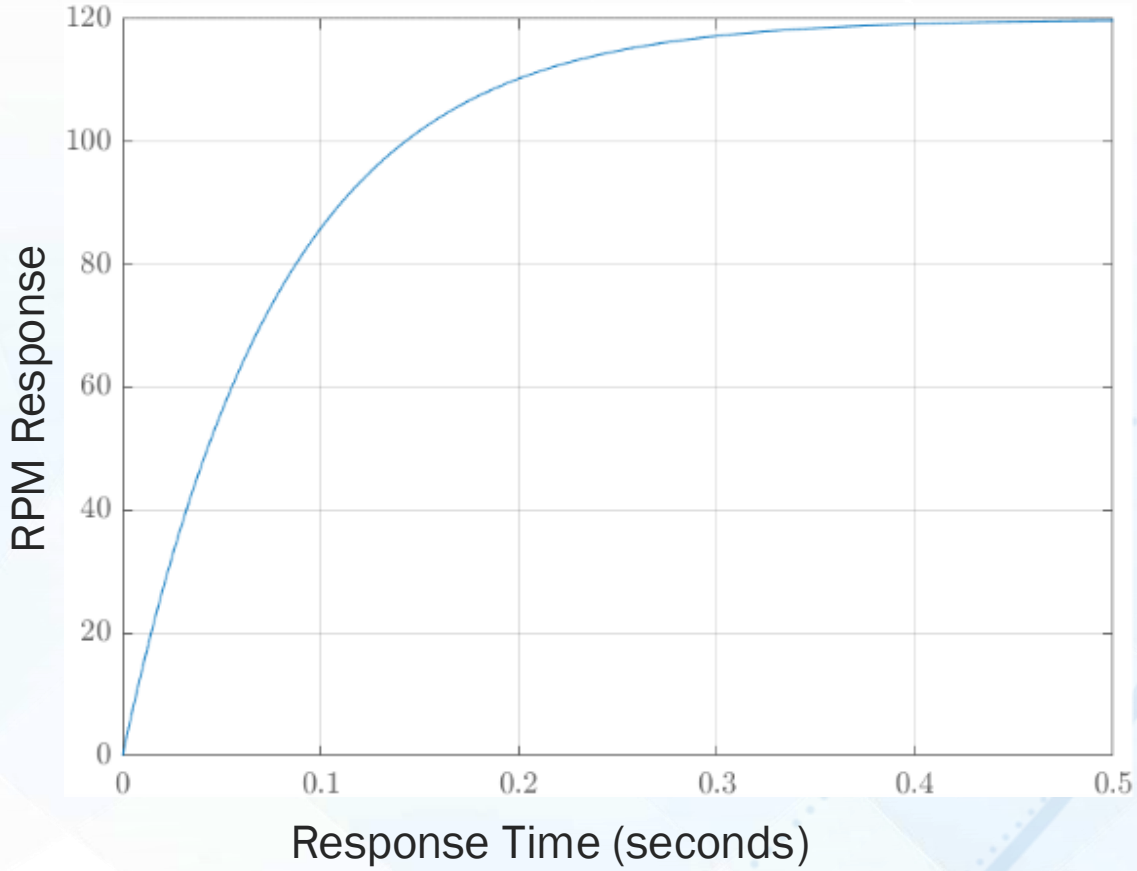
$$Y(s) = T(s) \cdot (240/s) = \frac{0.9693}{0.00493s + 1} \cdot (240/s)$$

Time Domain Step Response:

$$y(t) = 232.6 \cdot \left(1 - e^{\frac{-t}{0.00493}}\right)$$

Rise Time: 10.8 ms
98% Settling Time: 24.6 ms

Closed-Loop RPM Response Over Time

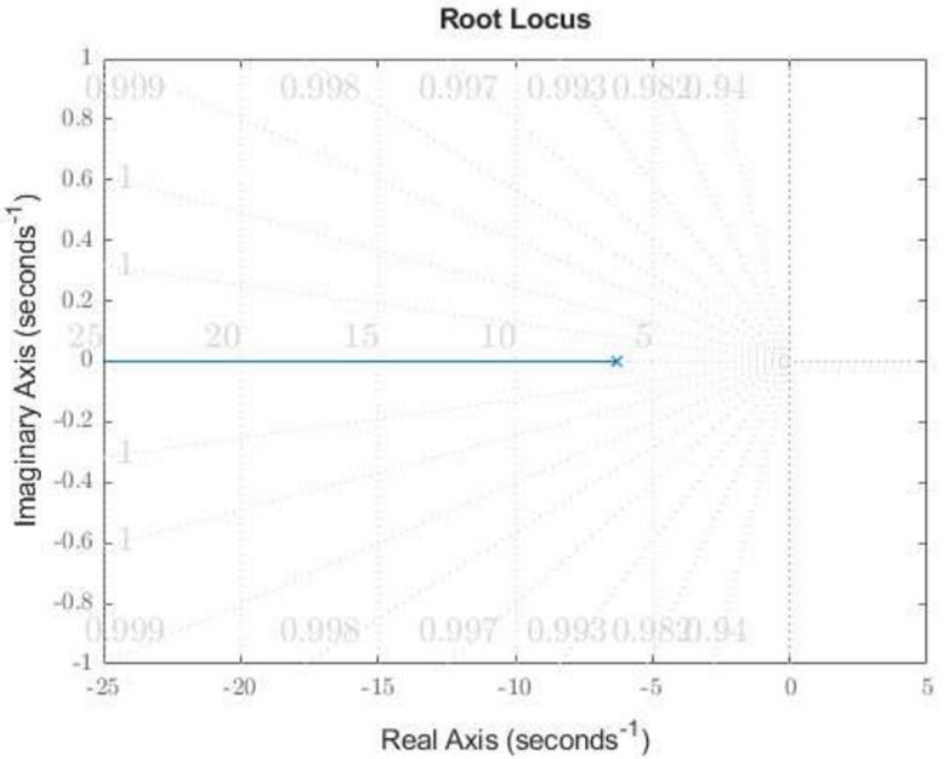


Root Locus and Proportional Controller Evaluation

Closed Loop Characteristic Equation:

$$0.159s + 1 + 31.2375K = 0$$

- Pole Originates at s = -6.29 rad/s at K=0
- Pole Stays Real and Negative for 0<K<∞
- Damping Ratio Stays ζ = 1
- System is Stable for any DC Gain
- Overshoot = 0
- Plant Type = 0
- Steady State Error Requires PI Controller



PID Controller Design

This section covers how the PID Controller was designed and optimized for this system

Goals and Objectives

The goal of this is to refine the closed-loop response by eliminating the steady state error and minimizing settling and rise time.

$$C(s) = 2 + \frac{15}{s} = \frac{2s + 15}{s}$$

$$G(s) = \frac{31.24}{0.159s + 1}$$

$$C(s)G(s) = \left(\frac{31.24}{0.159s + 1}\right)\left(\frac{2s + 15}{s}\right) = \left(\frac{62.48s + 468.8}{0.159s^2 + s}\right)$$

Final closed-loop transfer function:

$$\left(\frac{62.48s + 468.8}{0.159s^2 + 63.48s + 468.8}\right)$$

Results

The settling and rise times have improved minimally from the closed-loop times. The steady-state error introduced from the unity feedback has been fixed, and the system now settles at 240 RPM with less than 0.1% overshoot.

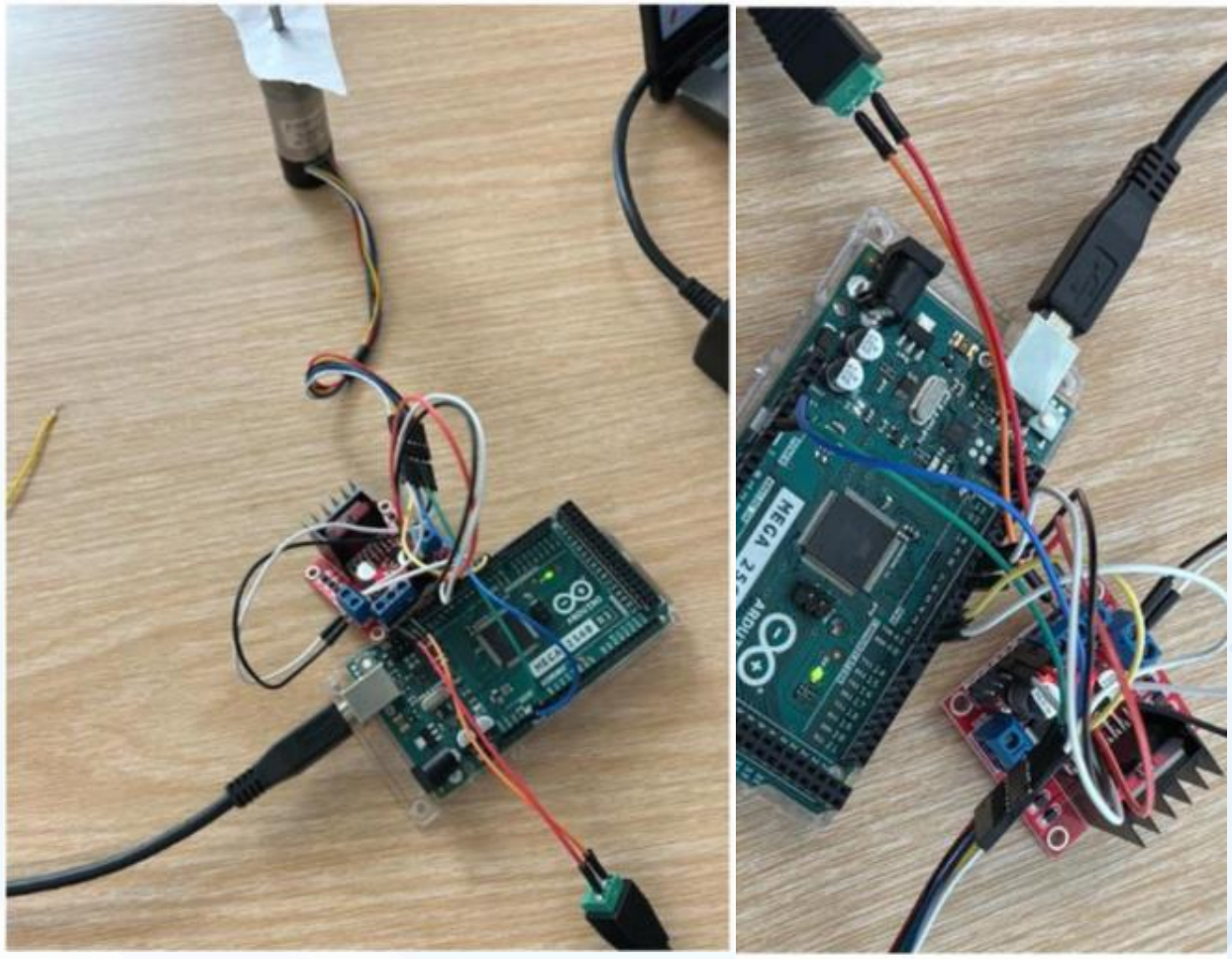
	Open-Loop	Closed-Loop	PID
Rise Time (s)	0.3493	0.1747	0.1695
Settling Time (s)	0.622	0.311	0.2909
Overshoot (%)	0	0	0.07
Peak Value (RPM)	239.84	119.92	240.17

Implementation of the Arduino

This section covers the implementation of the Arduino device as well as the comparison between the computed data for this transfer function and the actual output created from the Arduino IDE code.

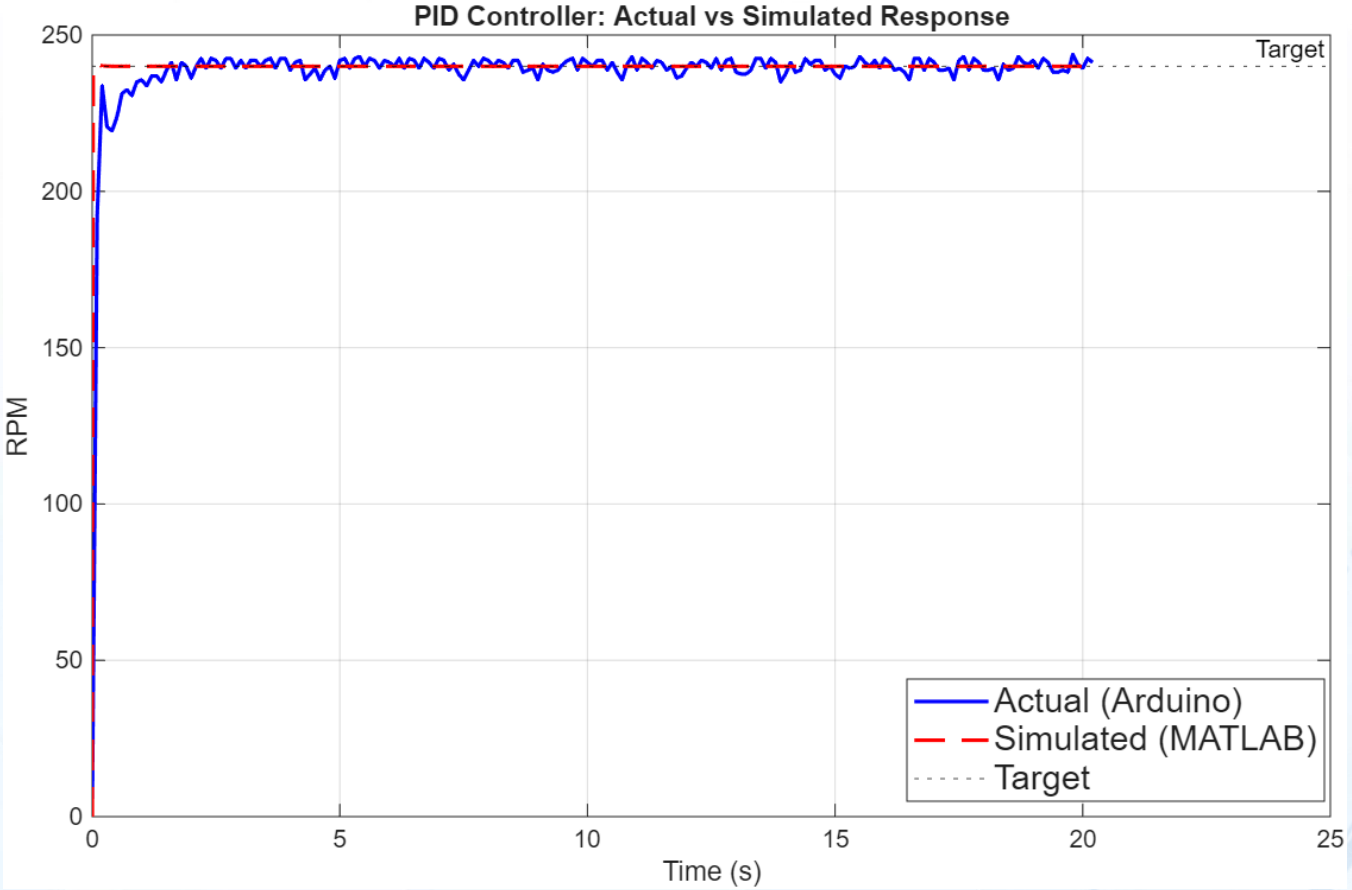
Circuit Setup for the Arduino

The setup of the circuit for this second portion was kept the same from the setup of project 1.



Comparison of Computed vs Actual response for the Arduino

The Arduino shows close agreement to computed data.



Conclusion

The unity feedback from the closed-loop response introduced significant steady-state error, which enhanced the need for a PI Controller. After designing this controller, the system settled at a desired 240 RPM with a settling time of 0.2909 seconds and a near-zero overshoot.

The experimental Arduino implementation confirmed a strong agreement between the simulated and measured responses, as the motor reached the target speed of 240 RPM.